

Inverted MAML Hardware Architecture

Discretized Time-Slice Stratification & Asymmetric Proprioceptive Clocking

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1. Executive Summary & Core Paradigm

In analog In-Memory Computing (AIMC) systems, non-deterministic charge leakage (voltage degradation governed by time-variant RC curves) constitutes the absolute physical barrier to digital-grade determinism. Conventional approaches implement complex, energy-expensive peripheral circuitry to actively fight this degradation, attempting to anchor analog nodes to static digital states.

This architecture flips the problem entirely. Instead of combating thermodynamic decay, the **Inverted Model-Agnostic Meta-Learning (Inverted MAML)** framework treats the entire non-linear leakage trajectory as a predictable, multi-dimensional computational feature. By shifting the objective from state enforcement to geometric adaptation, the system updates a single static initialization vector capable of navigating the continuous physical decay of the hardware substrate.

2. The Hidden Offset Bottleneck & Spatial Resolution

A primary architectural constraint of analog memory arrays is the asymmetry between digital processing speeds and analog programming intervals. The digital controller can only write **one and only one** static set of weights during a global refresh pulse. Once programmed, these analog values immediately begin drifting. Toward the asymptotic tail of the discharge window, all physical states drop significantly, generating a profound, state-dependent *unknown offset* relative to the intended digital model.

If the system executes standard Stochastic Gradient Descent (SGD) with Momentum using point-in-time derivatives, it incurs extreme **phase lag**. Sampling a single window near the beginning of the cycle ignores the tail-end collapse; sampling exclusively near the end overcompensates for the offset, destabilizing the next fresh state. This partial observability causes severe harmonic oscillations, model vibration, and tracking failure around the optimal parameter space.

3. Stratified Time-Slice Batching Topology

To systematically integrate out the non-linear nature of this physical leakage within the constraints of a single weight-update pulse, the continuous discharge window is decomposed into a structured **Stratified Batching** framework. The discharge cycle is partitioned into ten uniform temporal/voltage regions (Strata):

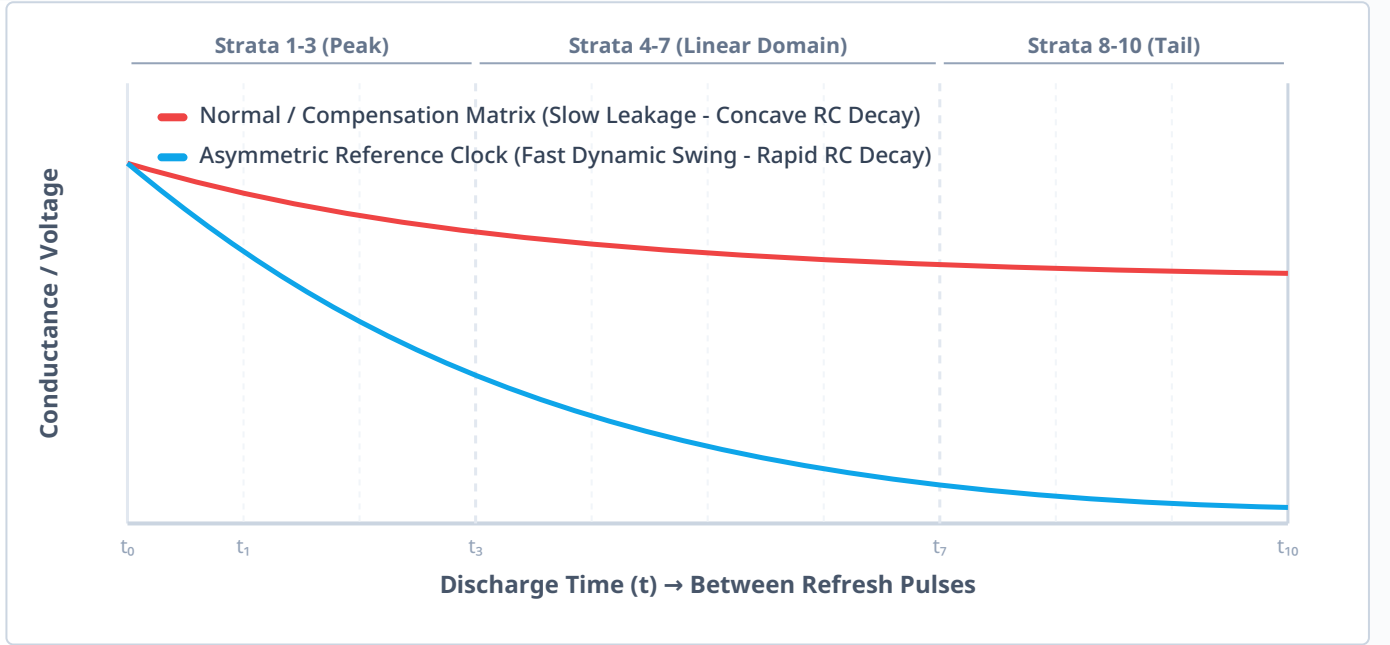


Figure 1: 10-Strata temporal partitioning mapping the rapid reference clock swing against slow concave compensation matrix degradation.

- **Stratum 1–3 (High Energy Domain):** Extracts gradients immediately following the digital refresh pulse, capturing sub-threshold initialization anomalies and peak parasitic leakage.
- **Stratum 4–7 (Linearized Decay Domain):** Evaluates the quasi-linear region of the RC curve where steady-state computational payloads spend the highest operational duration.
- **Stratum 8–10 (Asymptotic Tail Domain):** Captures the highly compressed, low-energy end of the discharge curve where cumulative offsets peak and approach the thermal noise floor boundaries.

During a single operational discharge cycle, the digital controller captures local differential derivatives $\nabla L(t_i)$ across all ten slices without modifying the analog state. Before the next refresh pulse, these snapshots are combined into a single stratified mini-batch update:

$$\nabla W = \frac{1}{10} \sum_{i=1}^{10} \nabla L(t_i)$$

By averaging derivatives across the entire physical curve, the algorithm executes a discrete integration over the time domain. The calculated update resolves the unknown offset by establishing a mathematically sound meta-learned tyngdpunkt (center of mass), yielding a starting parameter configuration optimized to minimize total error across the entire lifecycle of the subsequent pulse.

4. The Asymmetric Reference Clock & Symmetry Breaking

The core challenge in optimizing a completely analog neural matrix is *total covariance*: every component within the hybrid atomar triad (payload, compensation, and reference) is leaking simultaneously. If all components discharged at identical rates, the internal state transitions would become mathematically unobservable from external outputs. The local gradients would lose directional sign integrity, resulting in immediate model collapse.

This limitation is overridden by introducing deliberate **Asymmetric Discharge Rates**. The dedicated Reference cells are engineered with highly reduced RC constants, forcing them to discharge at an accelerated velocity relative to the payload and compensation weights (see *Figure 1*).

Mechanisms of the Asymmetric Proprioceptive Anchor:

- 1. Full Dynamic Swing as an Internal Clock:** The aggressive decay of the reference cells sweeps across a massive voltage spectrum in a short time, effectively converting a parasitic leakage behavior into an explicit, high-resolution analog clock pulse.
- 2. Physical Symmetry Breaking:** Because the reference signal drops fast, it drives a dominant, sharp gradient during the differential evaluation windows. This high signal-to-noise ratio completely drowns out the micro-fluctuations and minor drift occurring in the slower compensation weights.
- 3. Embedding into Layer 1 Input Space:** By routing these fast-decaying reference outputs directly as an explicit input feature into the first layer of the network, the model gains *proprioception* (self-awareness of its physical state). The Inverted MAML framework maps the instantaneous coordinate of this reference clock signal directly to the optimal compensation response required at that exact epoch of the discharge trajectory.

5. Architectural Conclusion

By structurally coupling Stratified Time-Slice Batching with an Asymmetric Reference Clock, the architecture achieves a stable, self-correcting equilibrium. 1-sample methods with momentum can establish a noisy, historical tracking center over many global cycles, but they are vulnerable to abrupt structural shifts (such as thermal spikes). The stratified batch architecture neutralizes the unknown offset natively within a single operational window, delivering digital determinism directly over an unshielded, dynamically blithed analog substrate.